

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

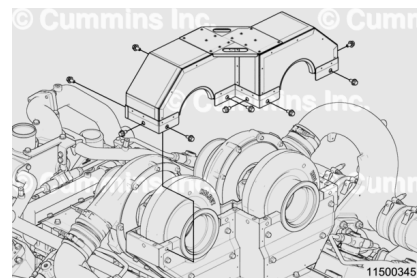
- Disconnect the batteries. See equipment manufacturer service information.

Remove

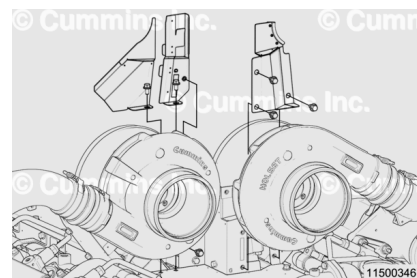
K38

Note : Rear out and front out configurations for heat shields are similar. The process below is for rear out, which can be used as a guideline for front out configurations.

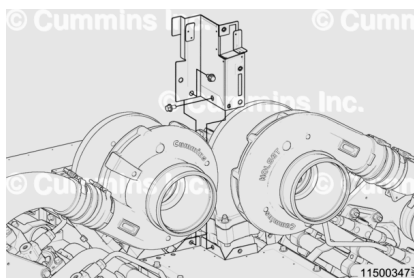
Remove the heat shield section from the top of the turbochargers by removing the nine capscrews around the shielding.



Remove three sections of heat shielding from the compressor side of the turbochargers.



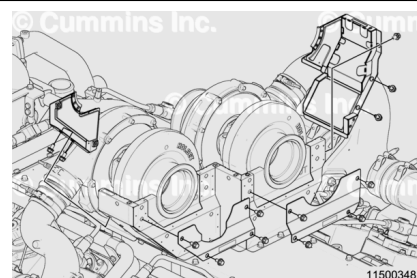
Remove the center section of heat shielding from the compressor side of the turbochargers.



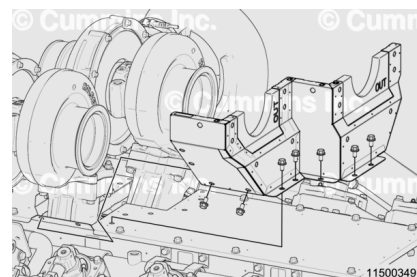
Remove the remaining section of heat shielding from the compressor side of the left bank turbocharger.

Remove the outboard section of heat shielding from the right bank turbocharger.

Remove the heat shield covers under the turbine outlets.

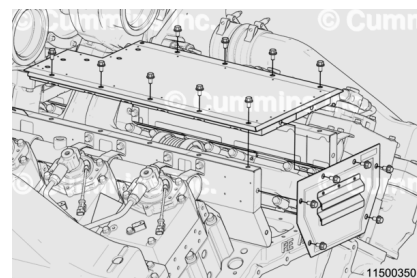


Remove the remaining section of heat shielding from the turbochargers.

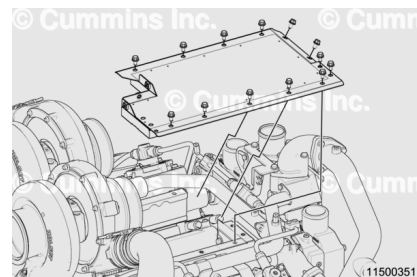


Remove the rear section of the exhaust manifold heat shielding.

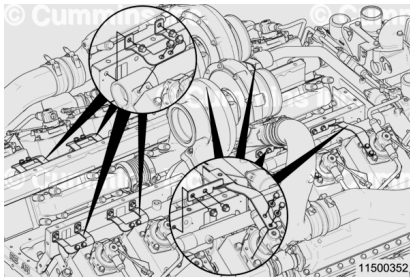
Remove the rear exhaust manifold cover.



Remove the front exhaust manifold cover.



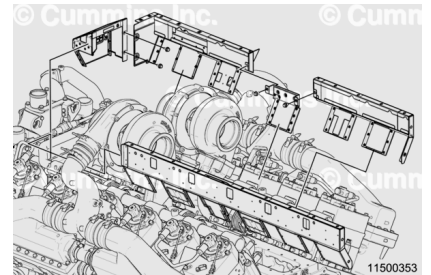
Remove the support brackets from the cylinder heads and side sections of heat shielding.



Remove the left bank, side section of heat shield from the exhaust manifold.

Remove the right bank, side sections of heat shield from the exhaust manifold.

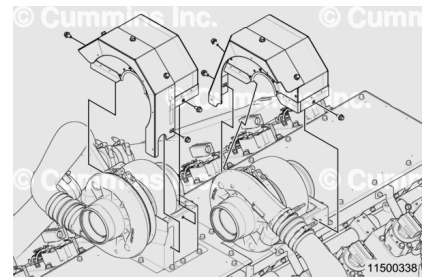
Remove the front section of heat shielding.



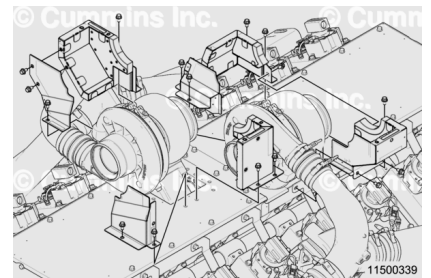
Single Stage

Remove the capscrews from the top section of the heat shields over the turbochargers.

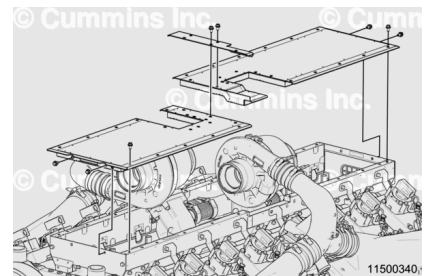
Remove the heat shields.



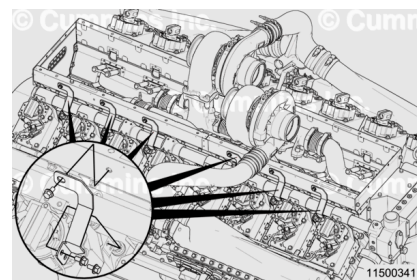
Remove capscrews and heat shielding around the bottom of the turbochargers.



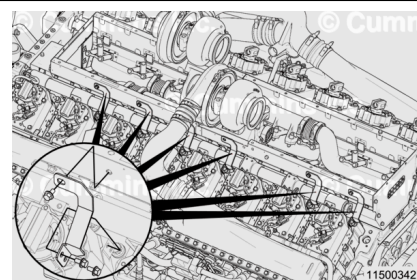
Remove capscrews and heat shielding over the exhaust manifolds.



Remove capscrews and brackets between the right bank rocker lever housings and exhaust manifold side shields.

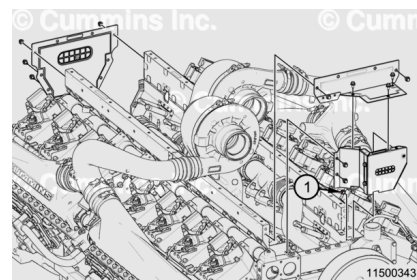


Remove capscrews and brackets between the left bank rocker lever housings and exhaust manifold side shields.

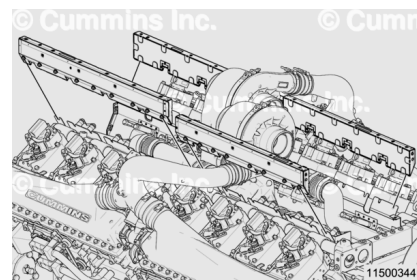


Remove capscrews and shielding from the front and rear of the engine.

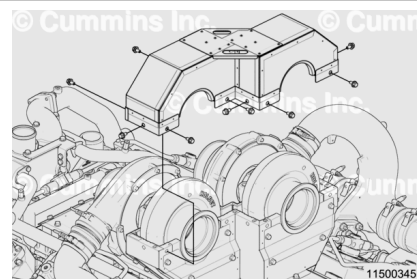
Be sure to remove the capscrew (1) in the front shielding that fastens from the rear.



Loosen the capscrews and slide the upper exhaust manifold side shields from the lower sections.



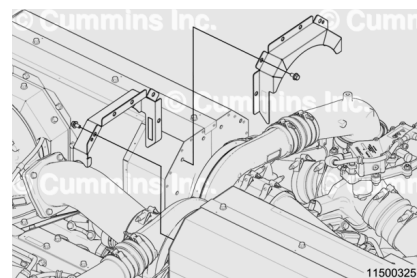
Remove the lower exhaust manifold side shields from the engine.



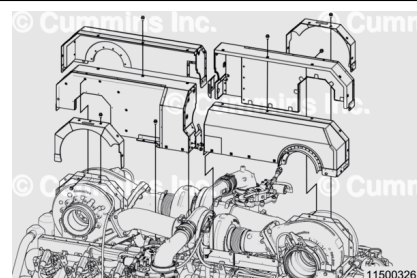
Two Stage

Remove the capscrews from the center section of the heat shield over the turbochargers.

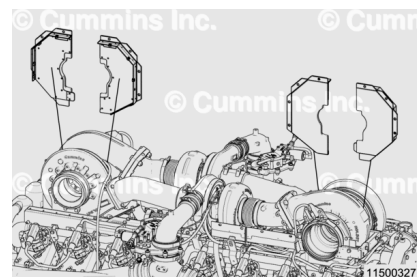
Remove the heat shield.



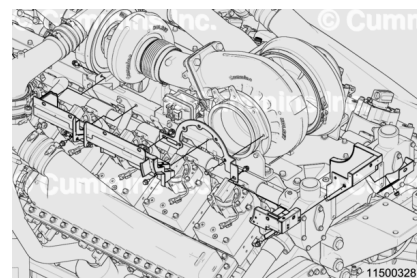
Remove the capscrews and top sections of the heat shields around the turbochargers.



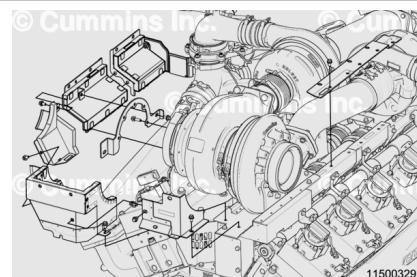
Remove the capscrews and shields from around the turbochargers on top of the set.



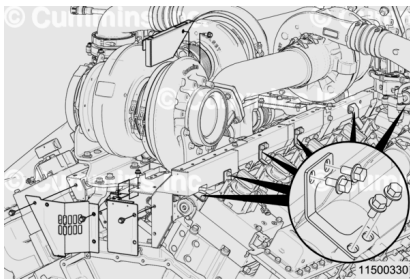
Remove the capscrews and heat shield working down the set.



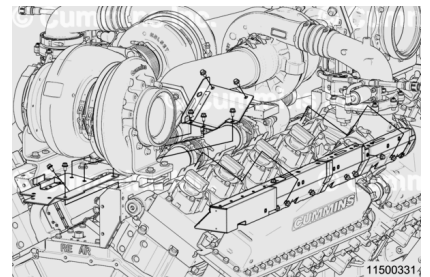
Remove the capscrews and brackets from the rocker covers.



Remove the capscrews and heat shield from the sides.



Remove the capscrews and sections of heat shield from around the sides of the engine.



Install

K38

Note : Rear out and front out configurations for heat shields are similar. The process below is for rear out, which can be used as a guideline for front out configurations.

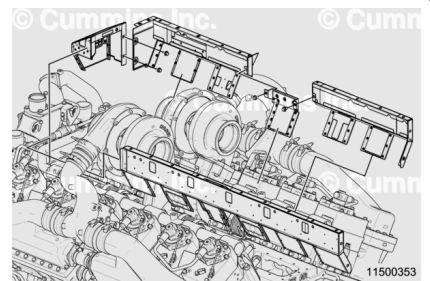
Ensure 5 mm to 10 mm clearance between heat shields and turbocharger oil drain hoses and coolant pipes.

Install the left bank, side section of heat shield to the exhaust manifold.

Install the right bank, side sections of heat shield to the exhaust manifold.

Install the front section of heat shielding. Tighten the two capscrews.

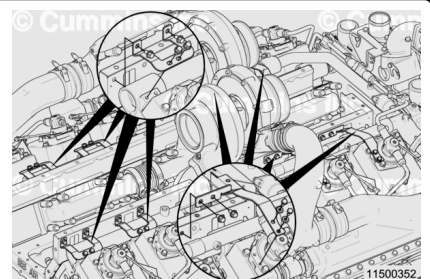
Torque Value: 23 n•m [204 in-lb]



Install the support brackets to the cylinder heads and rocker housings.

Torque Value:

Capscrews to Rocker Cover 27 n•m [239 in-lb]



Torque Value:

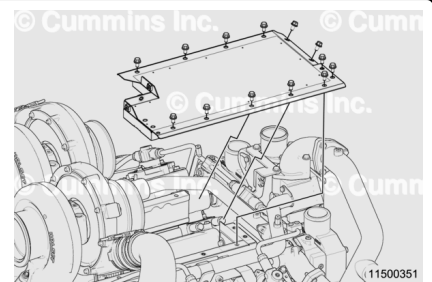
Capscrews to Rocker Housing 45 n•m [33 ft-lb]

Torque Value:

Capscrews to Heat Shield 38 n•m [28 ft-lb]

Install the front exhaust manifold cover.

Hand tighten the 13 capscrews.



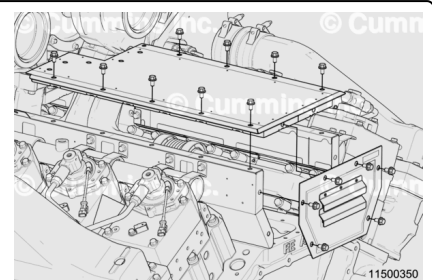
Install the rear exhaust manifold cover using two capscrews on either side.

Hand tighten the two capscrews.

Note : Do not install the top two capscrews.

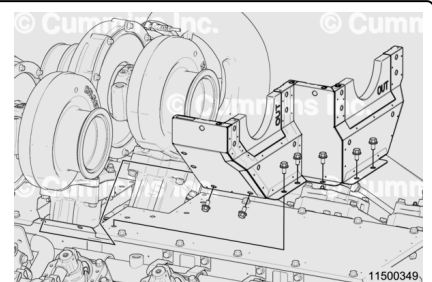
Install the rear section of the exhaust manifold heat shielding through the rear and side shields with nine capscrews.

Hand tighten the capscrews.



Install the first section of heat shielding around the turbochargers.

Hand tighten the capscrews.



Install the heat shield covers under the turbine outlets.

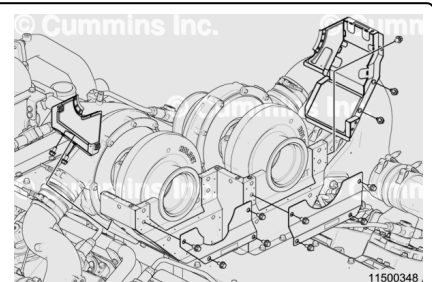
Hand tighten the capscrews.

Install the outboard section of heat shielding around the right bank turbocharger.

Hand tighten the capscrews.

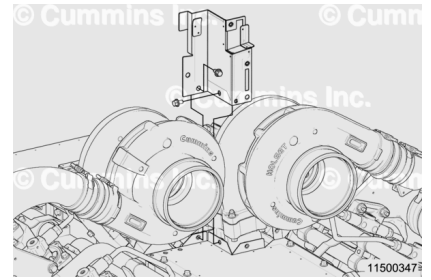
Install the remaining section of heat shielding from the compressor side of the left bank turbocharger.

Hand tighten the capscrews.



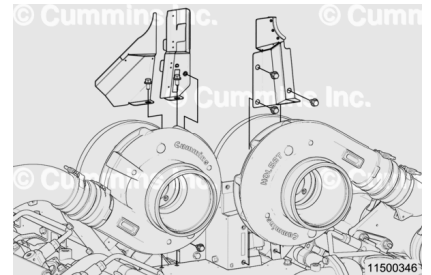
Install the center section of heat shielding on the compressor side of the turbochargers.

Hand tighten the capscrews.



Install the three sections of heat shielding on the compressor side of the turbochargers.

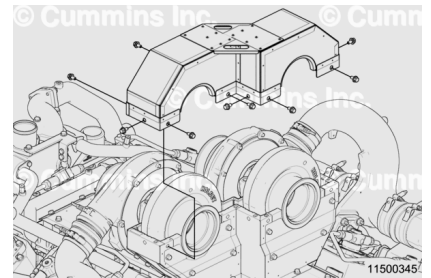
Hand tighten the capscrews.



Install the heat shield section from the top of the turbochargers using the nine capscrews around the shielding.

Hand tighten the capscrews.

Tighten all capscrews to the following torque unless otherwise stated above.



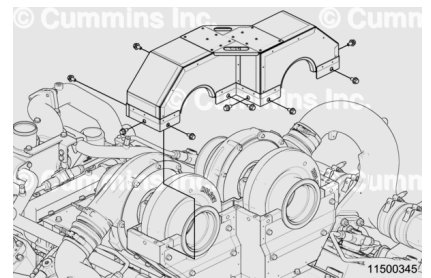
Torque Value: 27 n•m [239 in-lb]

Single Stage

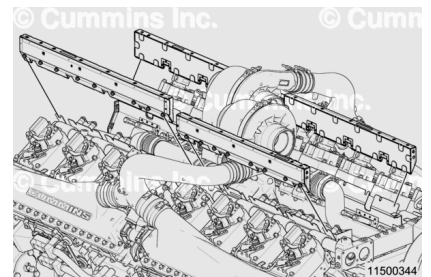
Note : For installation, reverse the removal procedure above. Hand tighten the capscrews until completely assembled unless otherwise stated.

Ensure 5 mm to 10 mm clearance between heat shields and turbocharger oil drain hoses and coolant pipes.

Install the lower exhaust manifold side shields to the engine.

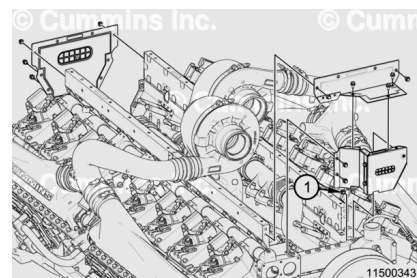


Slide the upper exhaust manifold side shields onto the lower sections and tighten the capscrews.



Install the capscrews and shielding to the front and rear of the engine.

Be sure to install the capscrew (1) in the front shielding that fastens from the rear.



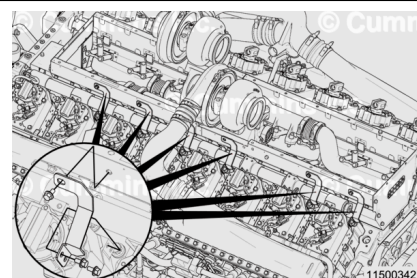
Install the brackets and capscrews between the left bank rocker lever housings and exhaust manifold side shields.

Tighten the capscrews into the rocker housing.

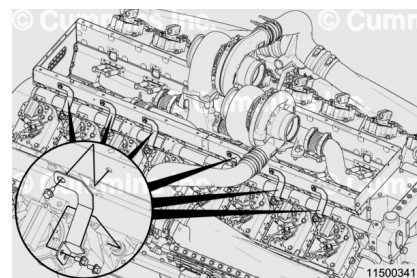
Torque Value: 45 n•m [33 ft-lb]

Tighten the capscrews into the heat shield.

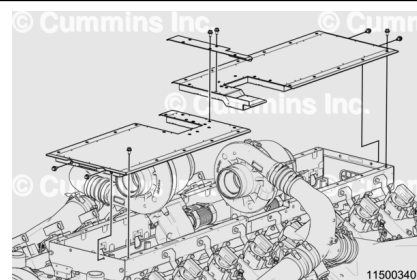
Torque Value: 38 n•m [28 ft-lb]



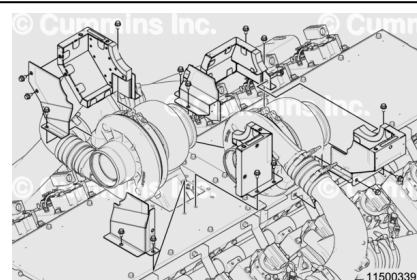
Install the brackets and capscrews between the right bank rocker lever housings and exhaust manifold side shields.



Install the capscrews and heat shielding over the exhaust manifolds.



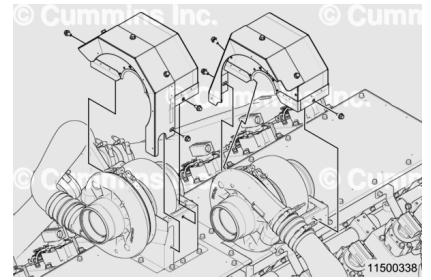
Install the heat shielding and capscrews around the bottom of the turbochargers.



Install the top sections of the heat shields over the turbochargers and tighten the capscrews.

Tighten the remaining capscrews.

Torque Value: 27 n•m [239 in-lb]

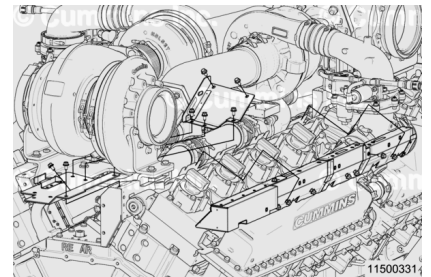


Two Stage

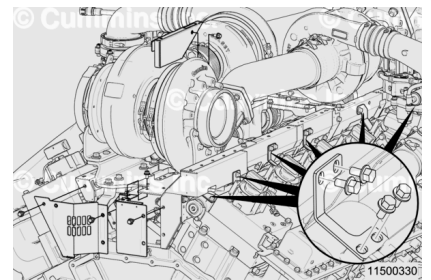
Note : For installation, reverse the removal procedure above. Hand tighten the capscrews until completely assembled unless otherwise stated.

Ensure 5 mm to 10 mm clearance between heat shields and turbocharger oil drain hoses and coolant pipes.

Install the sections of the heat shield and capscrews around the sides of the engine.



Install the heat shield and capscrews to the sides.



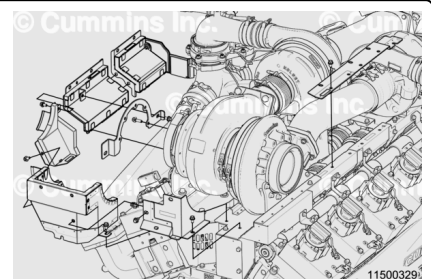
Install the brackets and capscrews to the rocker covers.

Tighten the capscrews into the rocker housing.

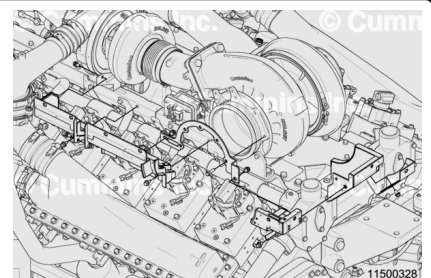
Torque Value: 45 n•m [33 ft-lb]

Tighten the capscrews into the heat shield.

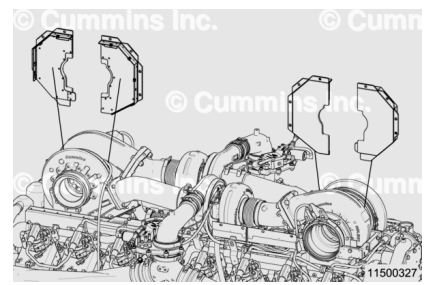
Torque Value: 38 n•m [28 ft-lb]



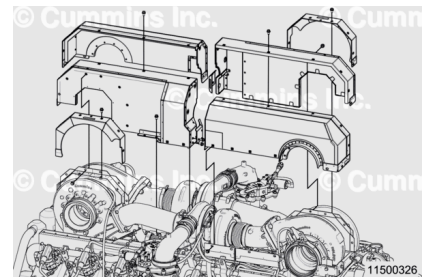
Working down the set, install the heat shield and capscrews.



Install the shields and capscrews around the turbochargers on top of the set.



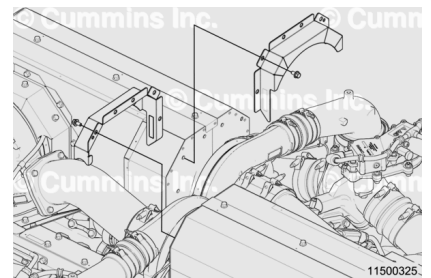
Install the top sections of the heat shields and capscrews around the turbochargers.



Install the heat shield and capscrews to the center section of the heat shield over the turbochargers.

Tighten the remaining capscrews.

Torque Value: 27 n•m [239 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. See equipment manufacturer service information.
- Start the engine and check for proper operation.

